Dear Editor,

I am submitting my manuscript titled "Testing the Fluorescence Advantage: Solar-Induced Fluorescence as an Early Indicator of Agricultural Drought Stress in Marathwada, Maharashtra" for consideration as a Research Article in the International Journal of Applied Earth Observation and Geoinformation.

This study tests whether Solar-Induced Fluorescence (SIF) picks up drought stress in crops earlier than the Normalized Difference Vegetation Index (NDVI), one of the main signals India's drought-declaration process and PMFBY crop-insurance assessment currently lean on. Using GOSIF v2 SIF data and MODIS NDVI across the 8 districts of Marathwada, Maharashtra, over 8 growing seasons, I found that SIF's decline came before NDVI's in most years, though not every year, and that the bigger gap is between satellite monitoring in general and the current official drought-declaration timeline.

I believe this paper fits the scope of your journal because it applies earth observation data directly to a real agricultural and policy problem, and tests the same question through several independent methods: threshold-crossing lag, cross-correlation, bootstrap confidence intervals, and spatial autocorrelation.

This manuscript is original and has not been published in, or submitted to, any other journal. An earlier version is available as a preprint on EarthArXiv and archived on Zenodo, as noted in the manuscript. I am the sole author and have approved this submission.

Thank you for considering my manuscript. I look forward to your response.

Sincerely, Sakshi D. Maske

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